

ESsense - An Electrostatic Sensor System for Simple, Contactless Motion Control in Games

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ABSTRACT

We showcase ESsense, a system of electrostatic (ES) sensors for alternative controllers, and its potential as a low-cost, easily-assembled motion detection system, to enable new interactions otherwise inaccessible to DIY controller makers. We characterize their most suitable interactions and demonstrate their ability to remap game inputs and recontextualise existing games with motion detection. The custom motion controllers provide button-based games with greater immersion than their original inputs, requiring motion that corresponds intuitively to the game's displayed actions and reactions. We also developed a game demo that showcases the potential of this technology to create new experiences. It gives the player the role of a music conductor, allowing them to control the tempo of a music track by swinging a baton in front of a conductor's stand. Given its ability to detect motion of any kind, we also discuss other possibilities for the system, such as in contexts beyond games.

CCS CONCEPTS

• **Hardware** → **Functional verification**; • **Human-centered computing** → **Interaction devices**.

KEYWORDS

games, alternative controls, sensors, motion-based, contactless

ACM Reference Format:

Liew Tse Chuan Joseph and Ng Keng Wei. 2022. ESsense - An Electrostatic Sensor System for Simple, Contactless Motion Control in Games. In *CHI Conference on Human Factors in Computing Systems Extended Abstracts (CHI '22 Extended Abstracts)*, April 29-May 5, 2022, New Orleans, LA, USA. ACM, New York, NY, USA, 6 pages. <https://doi.org/10.1145/3491101.3516484>

1 INTRODUCTION

Video game control schemes require large investments [16] in technologies, materials, and research. Furthermore, fragmented control schemes (e.g. the since-discontinued Nintendo Wii and its multiple controllers [3] for various games) become inflexible in contexts other than their specific use cases. This, among many other reasons, makes deep research and development into alternative controllers

less economically viable for companies to undertake, as their return on investment is very unpredictable (e.g. the motion-controlled Playstation Move won awards for immersive control [6] yet sold poorly [13]). This could disincentivize innovation in the field of video game controls.

It might be more efficient for users to build their own game controllers to suit the different contexts they might encounter, rather than rely on large game companies for controller innovations.

ES sensors can serve as a cost-effective and simple sensor technology that enables motion-based inputs. These can be produced easily, relatively quickly, and in unique configurations to suit different in-game interactions. Assembly and repair is also relatively simple and low-cost.

Our main motivation for this project started with a design project brief that entailed the exploration of triboelectric nanogenerators (TENG), which will be explained later, as a core technology. After researching into the topic further, and discovering that our region's humidity levels severely limits the function of TENG, we decided to employ ES sensors in final our works instead. This extended abstract is a unified collection based the sensor system we devised, showcasing the product of our research and experimentation with ES sensors in a more consistent fashion.

2 IMPLEMENTATION

Fundamentally, each ES sensor consists of a conductor, connected to a microcontroller, and a detached, electrostatically charged, moving dielectric material. When there is relative motion between the two materials without contact, the free electric charges within the conductor are disturbed, inducing current within the conductor [19] (Fig. 1). The induced current changes with distance and speed between the two materials. From our testing, setting the conductor as a stationary component and the dielectric as a moving component yielded the most consistent signal detection.

Past research [10] has shown that PTFE works as a good ES sensor material due to its low placement in the triboelectric series, meaning it readily gains and retains free electrons to attain a net negative electrostatic charge, which is ideal for the ESsense system.

The microcontroller can be programmed to track changes in current over time, and programmed to activate once a current threshold is crossed, allowing inputs to be sent to a game.

In environments like Singapore, the induced current is significantly lowered due to high humidity, which lowers electrostatic charge carried by all materials [14]. We had attempted to increase the signal passively by manipulating the conductor component, as shown in Fig. 2, but were unsuccessful.

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CHI '22 Extended Abstracts, April 29-May 5, 2022, New Orleans, LA, USA

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ACM ISBN 978-1-4503-9156-6/22/04.

<https://doi.org/10.1145/3491101.3516484>

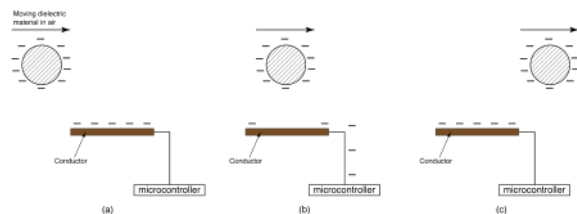


Figure 1: Illustration of the workings of an ES sensor. (a) A charged dielectric material moves toward a conductor. (b) The electric field of the conductor is disturbed, inducing a current within the conductor and its connecting wire. (c) As the dielectric moves away, the electric field in the conductor returns to its original state.



Figure 2: Attempts to boost signal through changing the shape of the conductor



Figure 3: Custom printed circuit board (PCB) for signal amplification and smoothing, connects up to two separate conductors

To work around this, a custom printed circuit board (PCB) as shown in Fig. 3 was used to amplify the generated signal and reduce electrical noise. Its effects are shown in Fig. 4.

For our use cases, digital, one-off inputs are detected. All our games' interactions are contactless, a unique capability afforded by ES sensors. When used in conjunction with each other, a series of sensors can be arranged to detect more elaborate actions than simply “on” and “off”, such that a number of physical quantities can be captured.

We built a demo (Fig. 5) showcasing some measurable quantities in the form of two ES sensor gates. When a dielectric material passes through the gates in succession, the time taken to pass through them can be captured, as well as their order of activation. With a

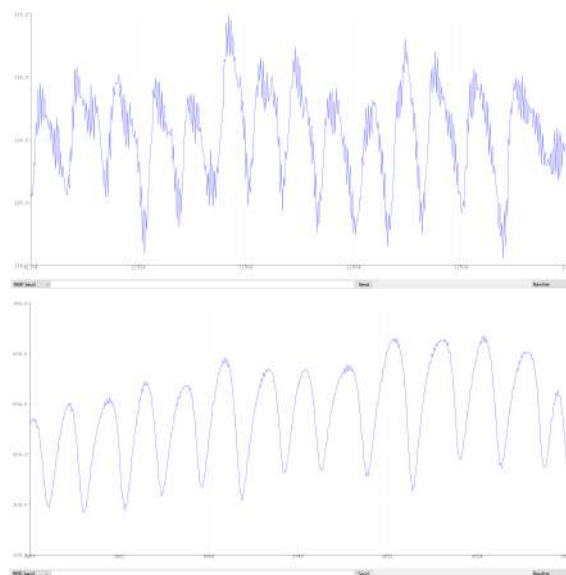


Figure 4: Moving a dielectric over a conductor without custom PCB attached (left) compared to attached (right), amplitude is multiplied approximately 12 times with PCB attached

known distance between the two gates, the speed of motion through them can be calculated [20].

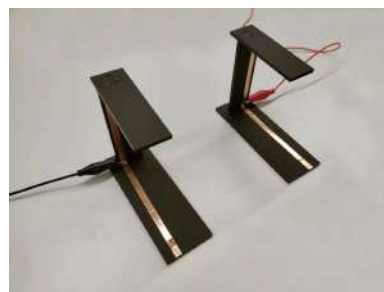


Figure 5: Demo of distance, speed, time, and direction identification

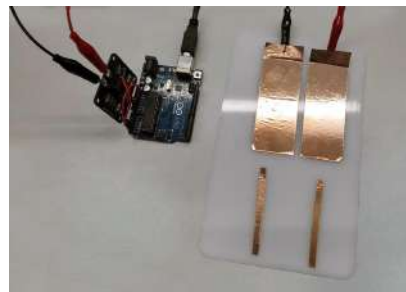


Figure 6: Our low-fidelity prototype to test some capabilities of ES sensors

The accuracy and specificity of the readings increases with the number of sensors used, for example, two-direction differentiation

is possible with two sensors, while three directions are possible with three.

With the core components of the system working, we set up pairs of copper tape with the custom PCB and an Arduino board to begin testing as shown in Fig. 6.

Our final prototype (Fig. 7) uses an Arduino Micro instead of an Arduino Uno, mainly because sending key presses to a computer makes use of an Arduino library that is accessible to the Micro model and not the Uno model, which were the only two models available to us. We placed the Arduino and the custom PCB inside a small case for easy connection to a computer and to an ES sensor for controlling games.

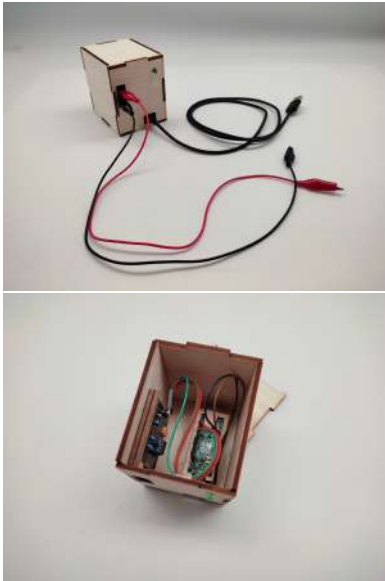


Figure 7: Our final prototype housing the Arduino Micro and custom PCB

3 GAME INTRODUCTION

We developed alternative controllers for two existing games by mapping the detected motion inputs to key presses, as well as built an original game using an interaction enabled by ES sensors.

3.1 Existing Computer Game - Lumberjack

In Lumberjack [4], the player controls a lumberjack to chop a tree while avoiding falling branches. The game is normally controlled by the left and right arrow keys on a computer for directional inputs.

Our custom controller (Fig. 8) requires the player to swing a model axe (moving dielectric) over a tree stump. The stump contains a pair of conductive pads (stationary conductors) that, when activated in succession, detect the direction of the player's swing. The directions are mapped to the left and right arrow key presses, allowing the game to react to the player's movements in a natural way. This remapped control scheme showcases the potential of the system to deliver a more convincing and immersive game experiences.



Figure 8: Screenshot of Lumberjack game (left), a web-based game from the Telegram messaging app and a custom controller (right) for it.

3.2 Remixed Game - Levitation Charm

A reskin of the popular smartphone game "Flappy Bird" [2] using Microsoft Makecode Arcade [5], the player acts as a wizard using a levitation charm on a feather, trying to avoid hitting the pillars of the wizarding school grounds as they move it around with their wand. The original game was normally controlled by tapping the smartphone's screen.

Our modification to the game (Fig. 9) requires the player to flick a wand (moving dielectric) over a tome (stationary conductor), doing which will send a spacebar key press to the computer, simulating the original controls of the Flappy Bird game. This game, with the afforded interaction allows players to feel like a wizard, floating the feather as if by magic. This reskinned game demonstrates the potential for recontextualising existing games and actions to suit the interactions made possible by ES sensors.

3.3 New Game Experience - Conducting Game

Our new game experience (Fig. 10) has the player take the role of a music conductor. The player swings a baton (moving dielectric) over a conductor's stand (stationary conductor) rhythmically, generating a tempo. This is passed into the computer and changes the tempo of a song played by the computer. If no swings are detected for a period of time, the game drops the track's tempo to 50bpm, a much lower tempo than most music tracks.

Indicator lights on the conductor's stand tell the player if their tempo is too slow, just right, or too fast. This encourages the player to adjust their tempo to match the original tempo of the song if they so choose.

This game makes use of an ES sensor connected to an Arduino microcontroller. Modulation of the music track is handled by p5js, a modified, web-based version of javascript. The p5js sketch receives tempo changes from the microcontroller through p5 Serial Control, a serial communication software made for p5js.

This game was made to bring the distant experience of music conducting to a wider audience, allowing players to better understand and appreciate the skill and focus of a music conductor with regards to maintaining tempo. Players will find that there is a constant tension between matching the track's original tempo and avoiding the distraction of the current playback tempo. This takes

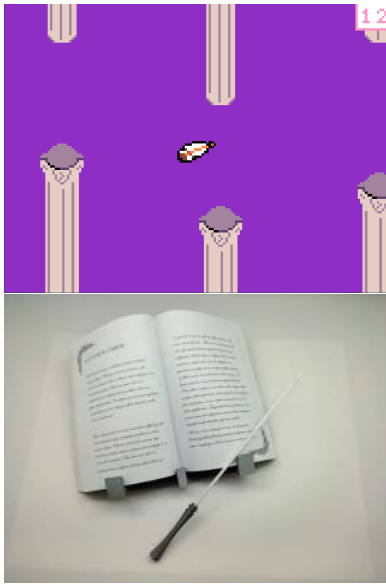


Figure 9: Screenshot of Levitation Charm (top), a reskin of Flappy Bird and a custom controller (bottom) for it.



Figure 10: Baton and conductor's stand

great focus, as completely ignoring the track's playback is not possible. Beyond that, it also serves as a fun way to manipulate music in real-time.

4 RELATED WORK

Currently, electrostatic sensors are used to measure and characterise the flow of multiphase particles, and to monitor the condition of mechanical machinery for industrial purposes.

The human body's electrostatic potential can change with movement bodily movements, allowing ES sensors to monitor human activity for medical and human-computer interaction (HCI) applications [19].

Non-contact applications of ES sensors have also been developed to monitor real-time hand position [17] and map hand movements to game controls [12].

Instead of tracking bodily movements, the ESsense system detects motion of a dielectric material, ideally polytetrafluoroethylene (PTFE). This enables users to easily build their own object-based customized controls with ES sensors easily and at low cost. While

existing sensor technologies exist for motion control, such technologies are not as easily repaired by an end user, requiring specialised parts.

Another technology that operates on similar principles to ES sensors is the Triboelectric nanogenerator (TENG) and triboelectric sensor (TES). These make use of electrostatic charge transfer by physical contact [18] to induce current, allowing for use in several applications [8]. There is limited electrostatic charge transfer as a result of high atmospheric humidity in Singapore [14], averaging at 75% [1], which limited charge transfer between parts of a TENG and TES. Furthermore, its requirement of physical contact between components made its unique applications rather limited, despite operating on similar principles to ESsense.

5 CHARACTERISING ES SENSORS

All sensors have different capabilities, methods of actuation, and complexity of construction. Oftentimes, multiple sensors are used in conjunction with each other to allow for accurate control in games (e.g. using a gyroscopes alongside accelerometers and geomagnetic sensors to capture a player's head movement [7]). Knowledge of these characteristics is important for anyone who might adopt the technology for their own use.

From our testing, ES sensors do not activate unless the dielectric moves in close proximity to the conductive material (<15cm of separation). They are also affected by electrical noise from nearby electrical devices [15]. This is a limitation that is best addressed through programming the microcontroller.

Despite being measured by the microcontroller as an analog signal, sustained input (i.e. a "press and hold" activation) cannot be measured. This is because constant relative motion is needed between the components to cause deflections.

The contactless nature of the interactions that we have built with the ES sensor also means they do not innately provide haptic feedback, which is sometimes important to digital game experiences. However, like other sensors, they can be used in conjunction with other systems to provide haptic feedback to the player.

Configuring several sensors within close proximity of each other proved unsuccessful, as the moving dielectric induces current, and therefore activates, all nearby sensors. Though electrostatic shielding of each conductive pad is theoretically possible, we were not able to isolate each sensor sufficiently to prevent accidental activation.

In summary, ES sensors can be loosely thought of as "contactless buttons" that detect the proximity and motion between particular objects, allowing for simple motion controls. Their configuration and environment are important factors to account for when building human-computer interactions with the technology.

6 INNOVATIVE INTERFACES

The novelty of ES sensors as a technology for alternative controllers mainly lies in the design opportunities it presents. These include enabling contactless interactions, their high configurability, and their low production and maintenance costs. Its capabilities are similar to that of a button, but enabling contactless, motion-based activation opens up many opportunities for Do-It-Yourself (DIY)

control setups for users that might not have been readily available previously.

The technology is easy to assemble from cheap components such as copper wire, aluminium foil, plastic bags, latex balloons, nylon sheets, etc., meaning configuring multiple ES sensors to enable different interactions is more cost-effective than other more sophisticated sensor technologies, while enabling unique interactions. By extension, its simple construction allows for easy and low-cost repair of controller parts as well. This is unlike ultrasound or gyroscopic sensors which require specialized parts, and are difficult to repair by end users.

We have built upon past submissions that focused on embodied controllers to provide unique, immersive experiences [9, 11, 21]. Our focus in this project was to showcase the potential of ES sensors as a simple platform to create or enhance game experiences through motion detection without contact. Despite being less accurate than existing technologies, its construction is also far simpler, lowering technical barriers for creating game controllers and experiences and allowing for more innovation in the field of alternative controllers.

We built another demo as shown in Fig. 11, featuring a pair of 3D printed ring accessories lined with PTFE sheets, and a wall-mounted conductive plate. We wanted to demonstrate that detection of hand movements such as waving, punching and slapping are possible. The actions can be detected by wearing the rings in different ways, as shown in Fig. 12.



Figure 11: An ES sensor set-up for detection of hand movement

7 CHALLENGES ENCOUNTERED

Being beginner coders, finding a solution to the large, fluctuating electrical noise was a significant challenge for us. The noise level differed depending on location, likely due to different surrounding electrical devices and wireless signals, as well as environmental humidity. Our best effort was to use ratios and percentages to smoothen the signal waveform to produce readable results for sensor activation.

Another challenge was adjusting the system's sensitivity for anyone to use without recalibration. From our testing, the amount of electrostatic charge carried by a person is unique and affects the sensitivity of the controller. To work around this, we adjusted our readings and thresholds to suit everyone who play-tested our system, and arrived at a threshold level that seemed to fit anyone regardless of the electrostatic charge they might carry.

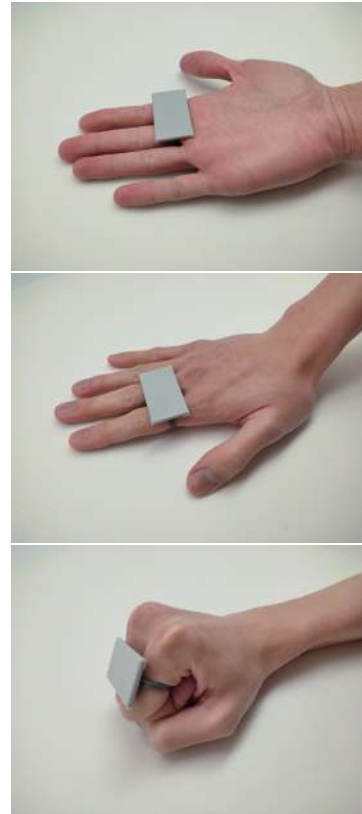


Figure 12: Different ways to wear and use the ring accessory

Lastly, we lacked the coding skills and knowledge to build our own games (integrating audio and visual computer experiences), thus requiring us to search online for games that either had interactions that were afforded by our system, or that could be easily modified to fit our system. That being said, this did lead to our discovery for the system's potential to bring new life to existing game experiences.

8 MOVING FORWARD

While still conjecture, we believe the ESsense system could be more intuitive to configure if presented in a recognizable medium such as a sticker set. This set could contain adhesive stickers with compatible ES sensor materials (e.g. copper, aluminium, PTFE, latex) that can be stuck to moving and stationary parts of DIY controllers. This could make most anything with moving parts a controller, opening up new possibilities for those who want to experiment with the technology more.

Applications beyond game controls could be created from this system too, digitising everyday inputs such as the direction of a door swing, the opened or closed state of a drawer (Fig. 13), or even a reminder system for wallets or medicines. Its could be used in physiotherapy too, as the custom controllers could be modified to suit one's exercise needs.

Optimising the ESsense system in terms of consistency in humid environments and shielding from adjacent sensors would be useful,

so as to overcome the issues mentioned above, improving reliability and variety of enabled interactions.



Figure 13: Visualizations of ES sensor configuration in daily applications

9 ACKNOWLEDGEMENTS

We thank Dr Yen Ching Chiuan for his guidance in focusing our core idea for this project, as well as offering us a workspace in NUS Smart Systems Institute to develop our prototypes and occasionally providing feedback on our work.

We thank Dr Clement Zheng for his guidance in preparing our paper for submission through teaching us about the LaTeX documentation system.

We thank Liu Xin for allowing us to use the amplifier PCB she had developed as part of her PhD research, and for explaining to us the fundamental workings and factors that affect TENG, TES, and ES in sensitivity and effectiveness. She also provided feedback and code samples which we modified for our sensors.

We thank Han Bo for his guidance with our project's technical requirements, including 3D printing, soldering and desoldering, as well as advice on our prototypes' usability and suggestions for new interactions.

We thank our seniors and friends for play-testing our prototypes and providing feedback on the user experience, as well as helping us with the linking of the Arduino to the computer through p5js for the Conducting Game.

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